



NAME: HET ACHARYA

COURSE: DIPLOMA IN CYBER SECURITY

DAY 5 – ASSEMBLE AND DISASSEMBLE A DESKTOP PC

Topic: Assemble and Disassemble a Desktop PC

MAIN COMPONENTS INSIDE A DESKTOP PC

Important Hardware Components

After opening a desktop PC cabinet, we can find several important components.

1. Motherboard

The motherboard is the main circuit board of the computer.
It connects different components and allows them to communicate with each other.

Motherboard = Main connection board

2. CPU

The CPU processes instructions and performs calculations.

CPU = Brain / Processing

3. RAM

RAM temporarily stores data that the computer is currently using.

RAM = Working memory

4. GPU

The GPU handles graphics and visual calculations.

GPU = Graphics

5. Storage

HDDs and SSDs are used to permanently store files, applications and the operating system.

Storage = Saves data

INTRODUCTION TO PC ASSEMBLY AND DISASSEMBLY

What is PC Assembly?

PC assembly means putting the different computer hardware components together and connecting them correctly. A desktop computer contains different components such as the motherboard, CPU, RAM, storage, GPU and power supply.

What is PC Disassembly?

PC disassembly means carefully removing the hardware components and disconnecting their cables from the computer.

Disassembly may be required for:

- Cleaning
- Repair
- Hardware replacement
- Upgrading components
- Troubleshooting

Simple Example

Think of a PC like a machine made from different parts.

Assembly → Put the parts together

Disassembly → Take the parts apart

Assembly = Put In

Disassembly = Take Out

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MORE PC COMPONENTS

6. Power Supply Unit – PSU

The PSU supplies electrical power to the different components of the computer.

It converts the incoming electrical power into suitable power for the PC.

PSU = Gives power

7. Cooling Fan

Fans help move air through the computer and remove heat from the internal components.

Fan = Moves air

8. PC Cabinet

The cabinet is the outer case that holds and protects the internal hardware.

It also provides mounting points and airflow space for the components.

Cabinet = Holds everything

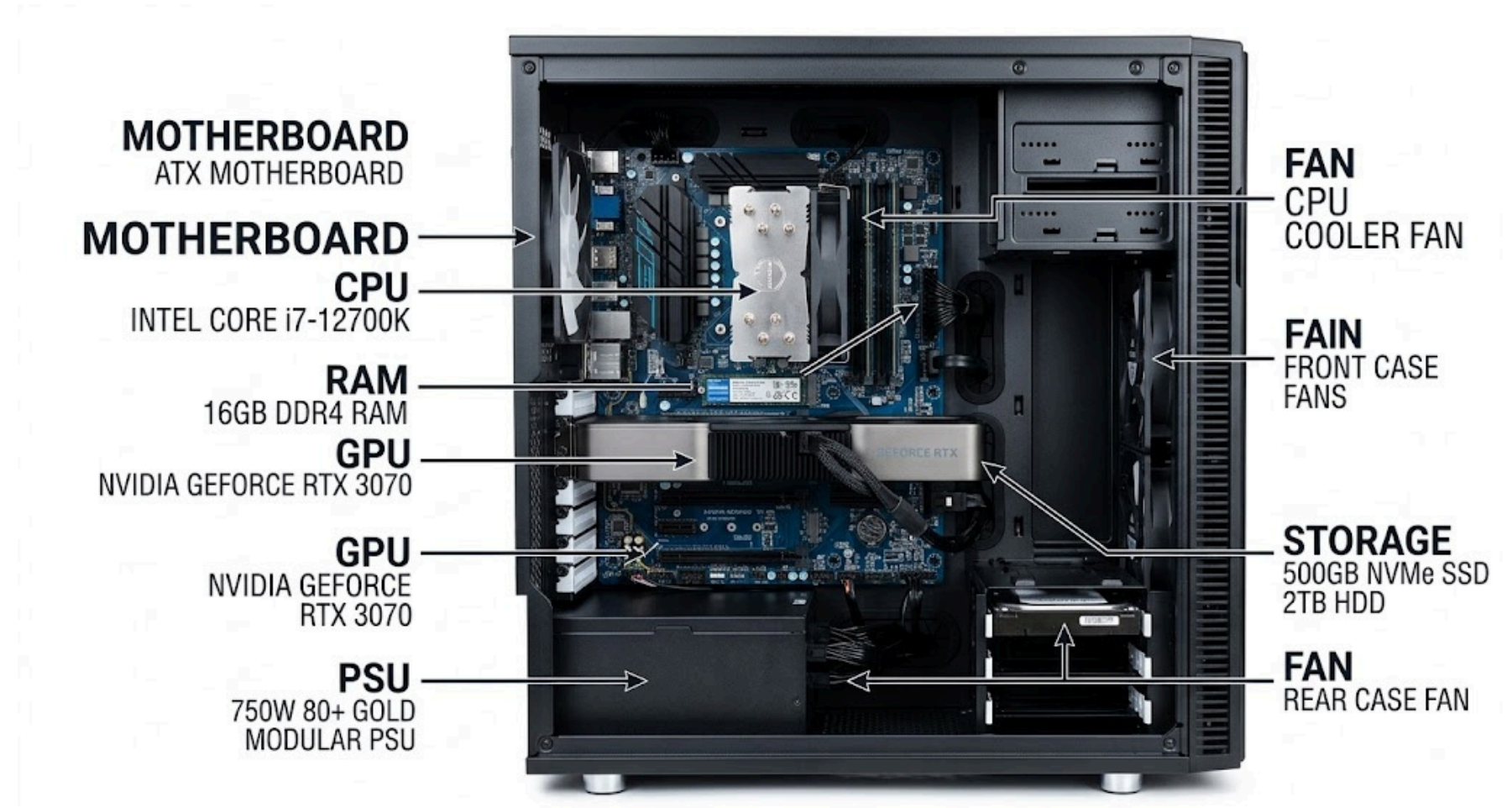


Figure 1: Main hardware components inside a desktop PC

SAFETY BEFORE DISASSEMBLY

Why is Safety Important?

Computer components are sensitive electronic devices. Incorrect handling can damage the hardware or create a safety risk.

Before opening the PC, some basic precautions should be followed.

Safety Rules

1. Shut Down the PC

Turn the computer off properly before starting.

2. Disconnect Power

Remove the power cable from the PC.

3. Remove External Connections

Disconnect unnecessary external cables and devices.

4. Use a Clean Workspace

Work on a clean, dry and stable surface.

5. Control Static Electricity

Use suitable anti-static precautions when handling internal components.

6. Handle Components Carefully

Hold components by their edges where practical and avoid unnecessary contact with electrical contacts.

Easy Safety Formula

POWER OFF → DISCONNECT → PREPARE → HANDLE CAREFULLY

TOOLS REQUIRED

Basic Tools

For basic desktop PC assembly and disassembly, we may need:

1. Screwdriver

Used for removing and tightening screws.

Memory:

Screwdriver = Screws

2. Anti-static Wrist Strap

It helps reduce the risk of static electricity when handling sensitive components.

Memory:

Wrist strap = Static protection

3. Small Container

Used for keeping screws safely in one place.

Memory:

Container = Keeps screws safe

4. Clean Workspace

A clean and organized area makes the process safer and easier.

5. PC Manual

A manual can help identify connectors, components and installation instructions.



Tools Checklist

- ☑Screwdriver
- ☑Anti-static wrist strap
- ☑ Small container
- ☑Clean workspace
- ☑ PC manual

Figure 2: Basic tools used for PC assembly and disassembly

DISASSEMBLY STEP 1: POWER OFF

Step 1 – Shut Down and Disconnect the PC

Before starting disassembly:

1. Shut down the operating system.
2. Turn off the computer.
3. Disconnect the power cable.
4. Disconnect other external cables if required.
5. Move the PC to a suitable workspace.

Why?

This reduces the risk of electrical problems and makes the system safer to work on.

Important Rule

Never start removing internal components while the PC is powered on.



PC powered off before disassembly

DISASSEMBLY STEP 2 AND 3 Step 2 – Remove the Side Panel

Before disconnecting anything, look carefully at the internal arrangement.
Take a clear photograph of the cables and components.



DISASSEMBLY STEP 4: DISCONNECT CABLES

Step 4 – Disconnect the Cables

The internal components are connected using different power and data cables.

Examples include:

- Motherboard power cable
- CPU power cable
- SATA cable
- GPU power cable
- Fan connections
- Front-panel connections

Important

Disconnect cables carefully.

Do not pull a cable by force.

Why?

Correctly disconnecting the cables prevents unnecessary damage to the connectors.

POWER + DATA + FAN CABLES

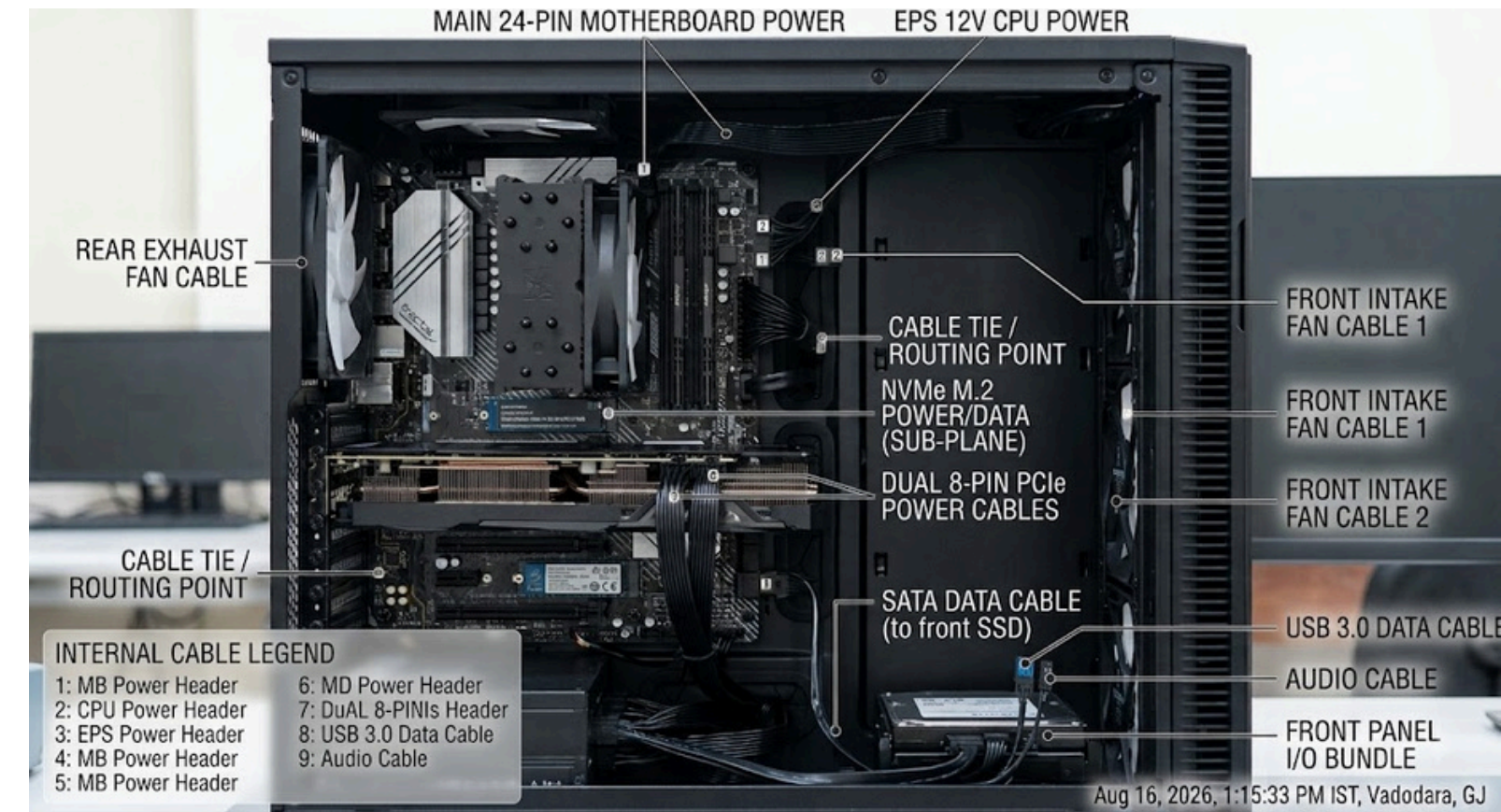


Figure 5: Internal power and data cables inside a desktop PC

DISASSEMBLY STEP 5: REMOVE RAM AND STORAGE

Removing RAM

To remove a RAM module:

1. Locate the RAM slot.
2. Release the slot clips.
3. Carefully remove the RAM module.
4. Hold it by the edges where possible.
5. Place it on a safe surface.

Easy memory:

RAM → Release → Remove → Protect

Removing Storage

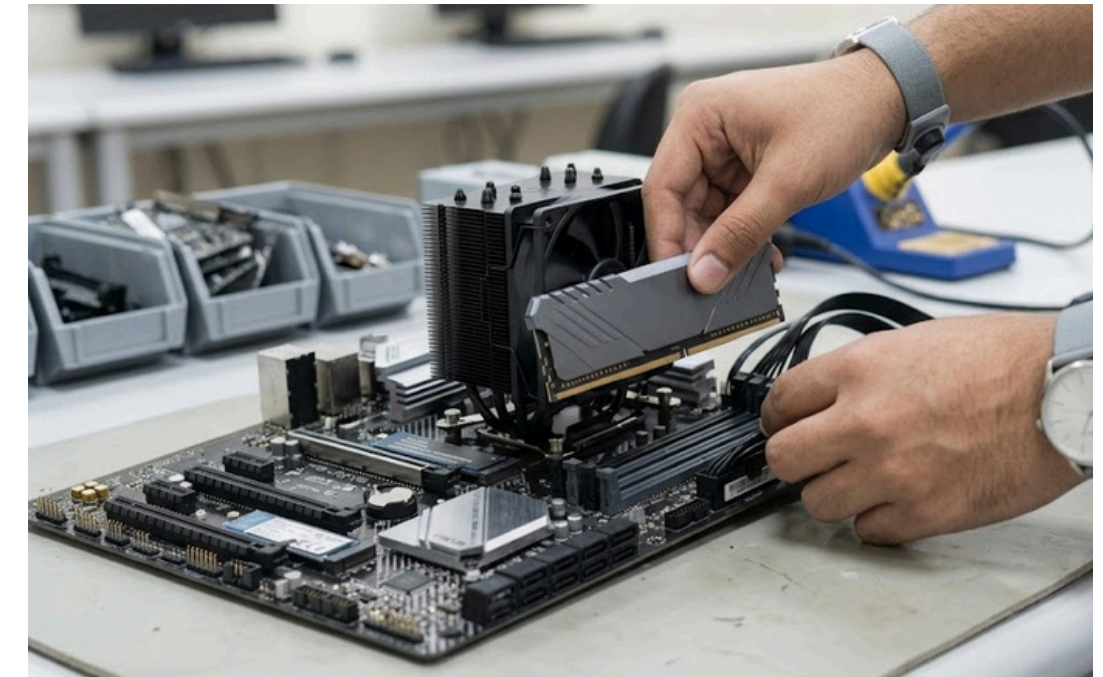
A computer may contain:

- HDD
- SATA SSD
- M.2 SSD

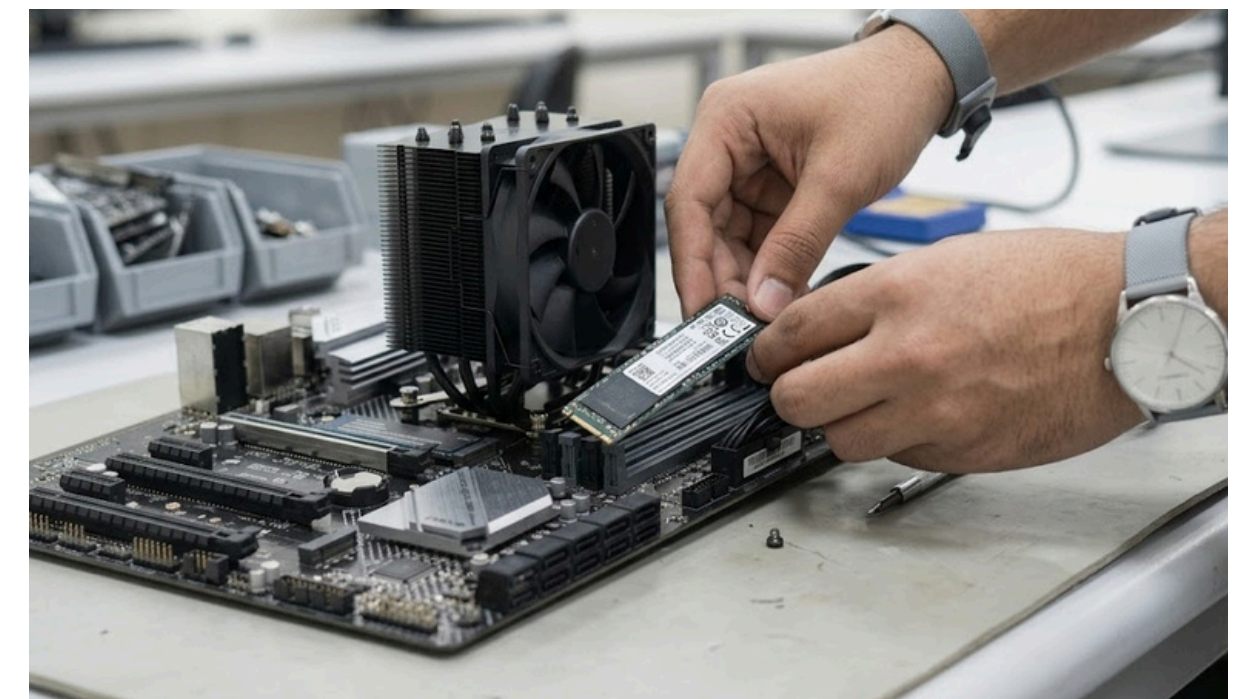
The exact removal process depends on the type of storage and how it is mounted.

Why remove storage?

Storage may need to be replaced, upgraded or inspected.



Removing RAM and storage components



DISASSEMBLY STEP 6: REMOVE GPU AND MOTHERBOARD

Removing the GPU

If the computer has a dedicated graphics card:

1. Disconnect its power cable if present.
2. Remove the case screw.
3. Release the PCIe slot mechanism.
4. Carefully remove the GPU.

Memory:

GPU → Power → Screw → Slot → Remove

Removing the Motherboard

After the required cables and components are disconnected:

1. Remove the motherboard mounting screws.
2. Check that no cable is still attached.
3. Carefully lift the motherboard.
4. Place it on a safe surface.

Important

Never force a motherboard out while a screw or cable is still attached.



Figure 7: Removing the GPU and motherboard during PC disassembly



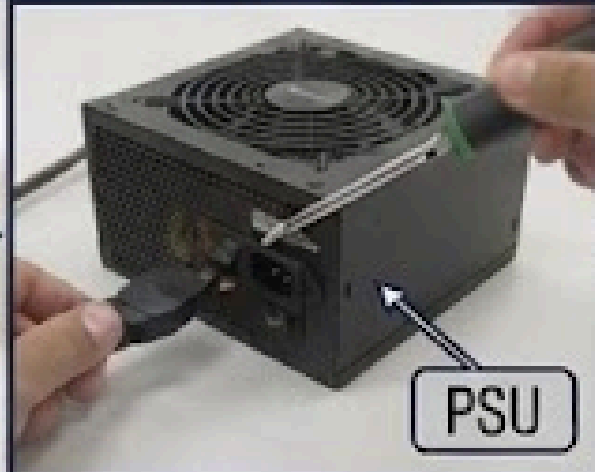
COMPLETE DESKTOP PC DISASSEMBLY PROCESS (9 STEPS)

STEP 1.



1. Power Off

STEP 2.



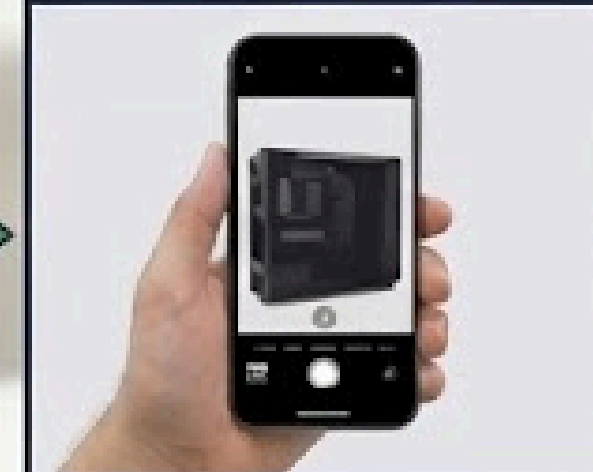
2. Disconnect Power

STEP 3.



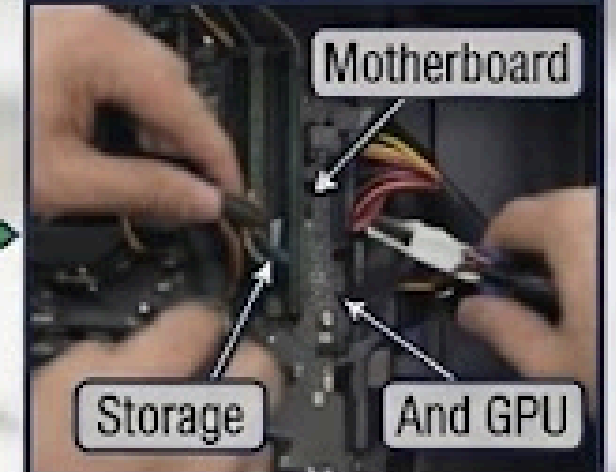
3. Remove Side Panel

STEP 4.



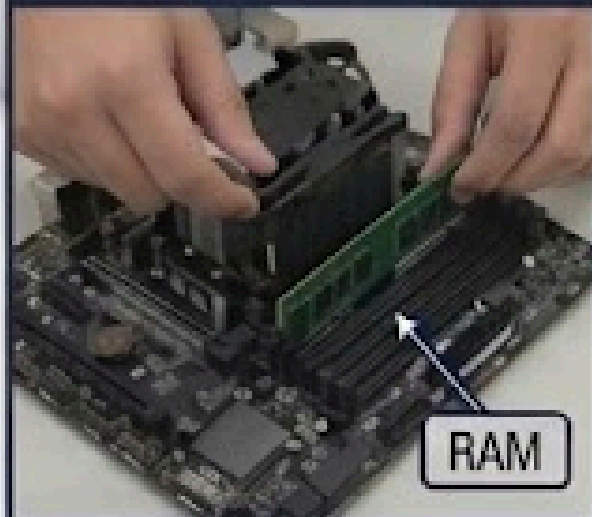
4. Take Reference Photo

STEP 5.



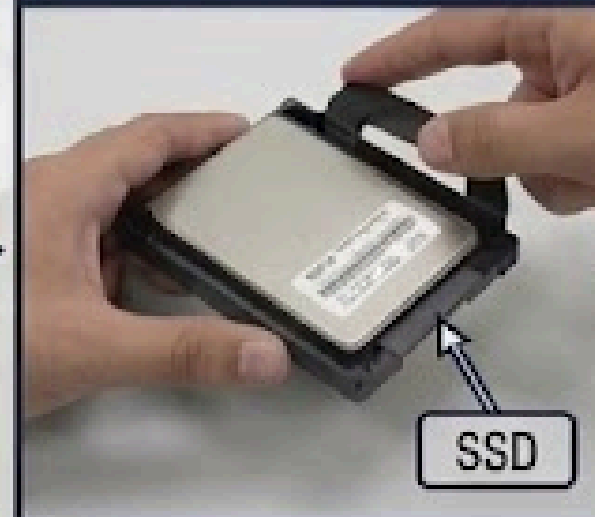
5. Disconnect Cables

STEP 6.



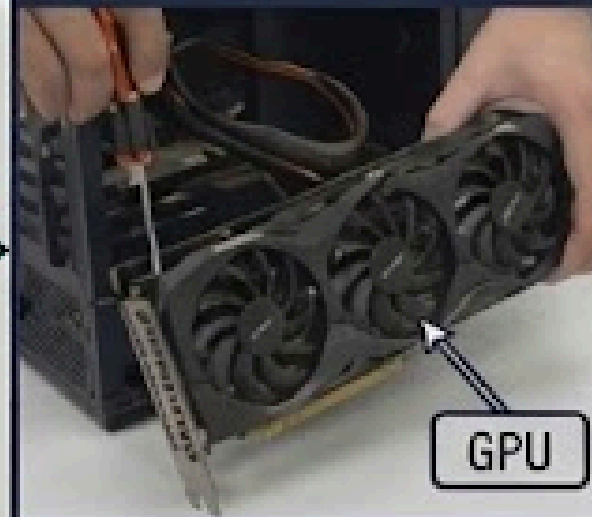
4. Remove RAM

STEP 7.



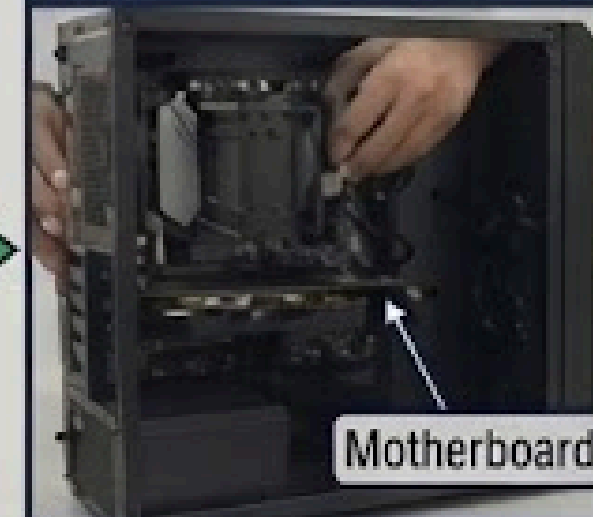
7. Remove Storage

STEP 8.

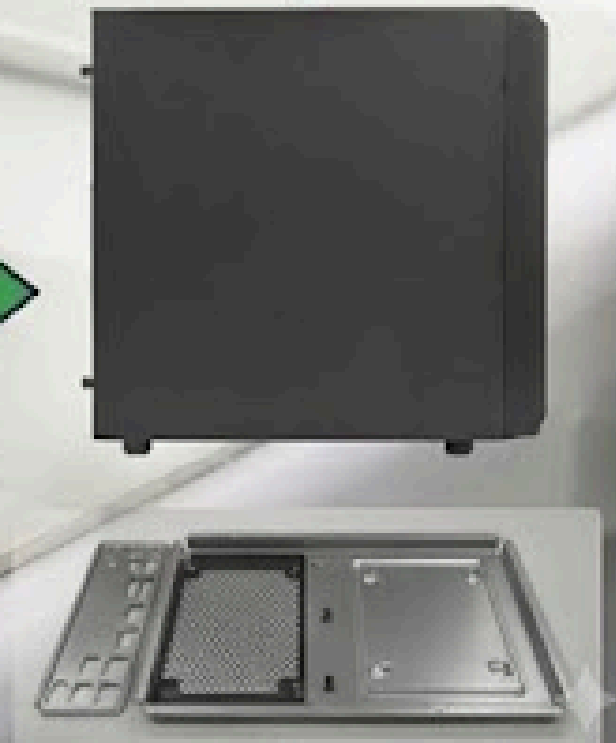


8. Remove GPU

STEP 9.



9. Remove Motherboard



PC REASSEMBLY

What is Reassembly?

Reassembly means installing the components back into the computer and reconnecting them correctly.

The process is basically the reverse of disassembly, but the installation order should still be planned carefully.

Main Steps

Motherboard → Storage → RAM → GPU → Cables → Final Check

Important Rule

Correct alignment is more important than force.

If a component does not fit correctly, check its orientation and connector instead of forcing it.

REASSEMBLY STEP 1-3

Step 1 – Install the Motherboard

Place the motherboard correctly on the mounting points and secure it with the appropriate screws.

Step 2 – Install Storage

Install the HDD, SATA SSD or M.2 SSD in its correct location.
Connect the required cables if it is a SATA drive.

Step 3 – Install RAM

Align the notch on the RAM module with the slot.
Press the module into position until it is properly secured.

Motherboard → Storage → RAM



Figure 8: Installing the motherboard, storage and RAM

REASSEMBLY STEP 4-6

Step 4 – Install GPU

If a dedicated GPU is used:

Place it into the correct PCIe slot.

Secure it to the case.

Connect its power cable if required.

Step 5 – Reconnect Cables

Reconnect the necessary cables.

Common connections:

Motherboard power

CPU power

Storage

GPU power

Fan connections

Front-panel connections

Step 6 – Final Inspection

Before turning on the PC, check:

RAM is properly seated

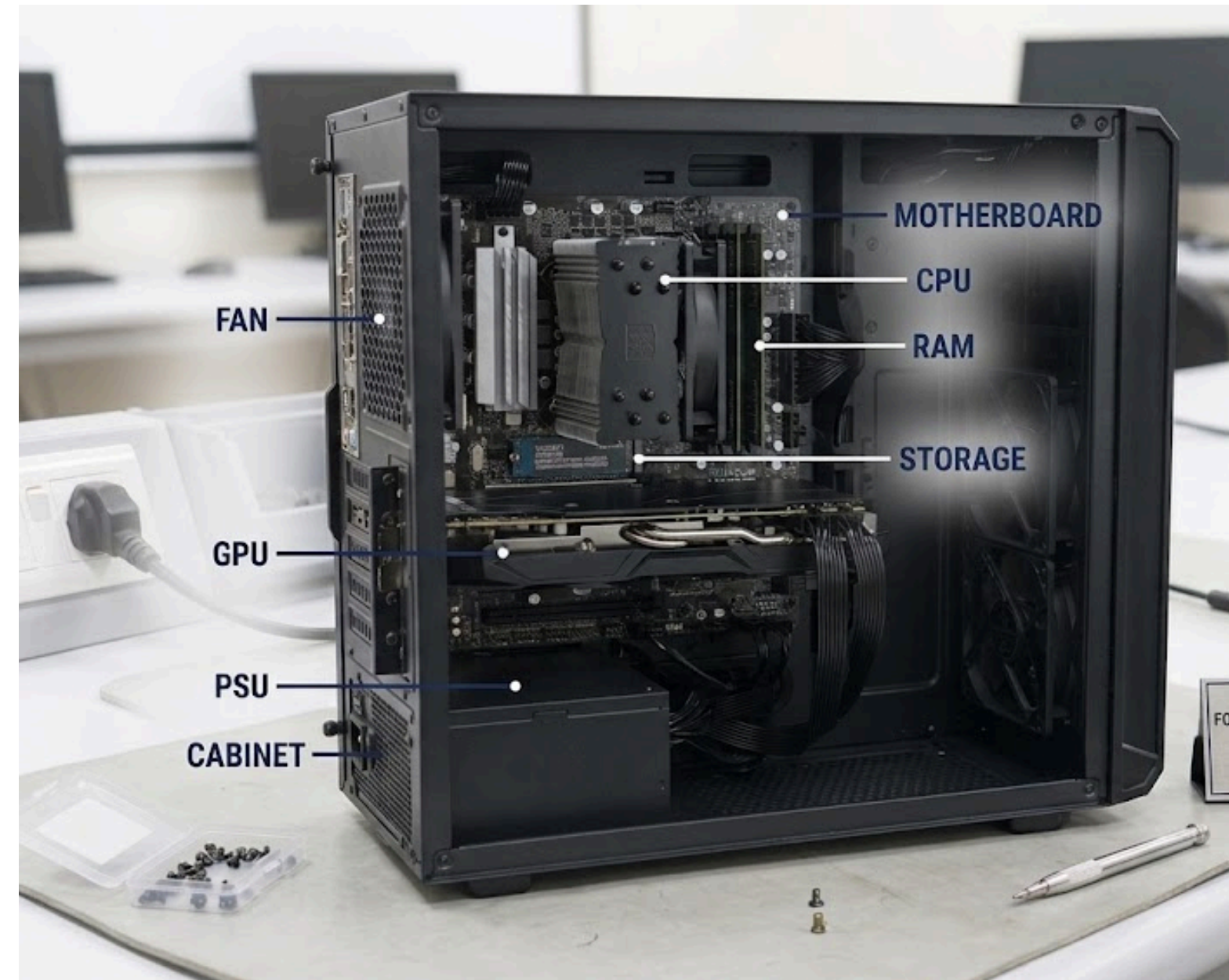
Storage is connected

GPU is secure

Power cables are connected

Fans are connected

No loose screws are inside the case



FINAL OBSERVATION & CONCLUSION

FINAL CHECK

Before powering on the PC, I checked:

- ☑ RAM is properly installed
- ☑ Storage is connected
- ☑ GPU is secured
- ☑ Power cables are connected
- ☑ Fans are connected
- ☑ No loose screws are inside the cabinet

MY PRACTICAL OBSERVATION

During this practical, I learned how the major hardware components are arranged and connected inside a desktop PC. I understood that components must be handled carefully and installed in the correct position.

I also learned that taking a photo before disassembly can help during reassembly.

CONCLUSION

PC assembly and disassembly helped me understand the practical structure of a desktop computer.

I learned the basic process of removing and reinstalling RAM, storage, GPU, motherboard and cables. I also understood that safety, correct alignment and proper connections are important when working with computer hardware.

“A PC works as one system when every component is correctly placed, connected and checked.”